cron—of Covid so far, Omicron has resulted in the least severe infection and symptoms but it has also been the most transmissible. "One reason we could have more sub-variants of Omicron is the way in which it has spread and continues to spread, giving it opportunities to mutate," says Dr Rakesh Mishra of the Tata Institute for Genetics and Society in Bengaluru.

Indeed, the latest sub-variants of Omicron—BA.4 and BA.5—are already being held responsible for the current spike. First spotted in South Africa in April and linked to the nation's subsequent Covid wave, they have been detected in dozens of countries since and account for nearly 25 per cent of all cases in the US. In India, Maharashtra has recorded the highest number of cases—49—of the two variants. According to NCDC or National Centre for Disease Control estimates, BA.4 and BA.5 grew from causing about one per cent of new Covid cases in India in the first week of May to causing 22 per cent of new cases in the week ending June 11.

While the disease they cause is mild, the sub-variants have unique mutations, including changes labelled L452R and F486V in its spike protein, that make it more infectious and capable of overcoming immune responses from previous infections and vaccines. And that is where the new concern lies—how effective is the current lot of vaccines against new variants and if a booster dose really helps. Dr N.K. Arora, head of the Covid-19 working group at the National Technical Advisory Group on Immunisation (NTAGI), is unequivocal in his advice: "Besides wearing masks and social distancing, we need to take booster shots, which build immunity and reduce risk of severe disease."



"Covid is still around and non-adherence to Covid-appropriate behaviour and low uptake of booster doses have contributed to the spread"

– Dr Randeep Guleria
Director, AIIMS, New Delhi



TOP SHIELD
A healthcare worker administers a Covid vaccine in Gurugram

THE BOOSTER DEBATE

So far, less than five per cent adults in India have taken a booster dose. Other than the confusion about which booster to take, many are wondering if it is worth taking at all. Dr Pavithra Venkatagopalan, director, Covid task force, Rotary Club of Madras Next Gen, cannot understand it. "If people don't think twice before taking the chickenpox vaccine, why should there be concern over Covid shots?" she asks. LNJP medical director Suresh Kumar is emphatic. "Boosters are essential to replenish your immunity against Covid-19," he says. Dr T. Chandrashekhar, internal medicine specialist at Fortis Hospital, Vashi, Mumbai, says that for those with comorbidities or the elderly, taking a booster is essential.

Many of the doubts seem to stem from studies that indicate vaccines to be largely ineffective against the new Omicron sub-variants. A new study by the Harvard Medical School claims that BA.4 and BA.5 appear to escape antibody responses among both those who have had an infection and those who have been fully vaccinated and boosted. The study found that the neutralising antibody levels were lower by a factor of 6.4 against BA.1, 7 against BA.2, 14.1 against BA.2.12.1 and 21 against BA.4 or BA.5. However, it did note that vaccines still provide protection against severe Covid and death. Studies have shown that even those with hybrid immunity—a result of vaccination and previous infection with Omicron—produce antibodies that struggle to stall BA.4 and BA.5. On June 28, its Food and Drug Administration or FDA recommended that the US update its boosters by the end of the year to target Omicron